Pareto-Optimal Redistricting: Transparent Trade-offs via Multi-Objective Genetic Algorithms

B.26 — Algorithm Design / Multi-Objective Optimisation

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Abstract

All current redistricting optimisation uses *weighted-sum objectives*: practitioners minimise $w_1 \cdot \mathrm{EC} + w_2 \cdot \text{partisan\_deviation} + w_3 \cdot \mathrm{VRA}_{\text{deficit}}$, tuning weights until a desired plan emerges. This is opaque: the weight choice determines the outcome, yet the relationship between weights and plans is non-linear and non-transparent.

We introduce Pareto-optimal redistricting, which instead returns a *frontier* of plans — the NSGA-II Pareto front — where no plan is strictly better than any other on all three objectives simultaneously. Practitioners choose from the frontier with full visibility of trade-offs: "this plan gives more compact districts at the cost of two fewer proportionally-owed seats."

The three objectives are (1) EC: graph edge cuts (compactness), (2) D_{seats}: distance from proportional seat allocation (partisan neutrality), and (3) $\mathrm{VRA}_{\text{deficit}}$: minority VAP shortfall below 50% in formerly majority-minority districts (VRA compliance). All three are minimised.

The central legal application is the *enacted-plan dominance test*: if the NSGA-II frontier contains a plan that strictly beats the enacted plan on all three objectives

simultaneously, the legislature chose a provably worse option on its own stated criteria. This is direct, non-statistical evidence of gerrymandering, distinct in character from ensemble-based arguments.

We present NSGA-II adapted for redistricting plans as chromosomes, with a ReCom-style crossover operator and a single boundary-tract flip mutation. Empirical results on NC ($k = 14$, 14 congressional districts) and WI ($k = 8$) demonstrate meaningful Pareto frontiers of 10–30 plans and characterise enacted-plan dominance for both states.

D__seats note: With NC $k = 14$, D_{seats} takes only $\sim$14 discrete values. The Pareto ordering may then collapse to a two-dimensional EC–$\text{VRA}_{\text{deficit}}$ frontier when many plans share the same D_{seats} value; this is acknowledged in output metadata (`d_seats_discrete: true`). Phase 2 will introduce vote margin as a continuous proxy.

Contents

1. Introduction

1.1. The Weighted-Sum Problem

Every redistricting optimisation tool in current use — including every algorithm in this platform — employs a *weighted-sum objective*:

$$\min\ w_1 \cdot \mathrm{EC}(\text{plan}) \ +\ w_2 \cdot \text{partisan\_deviation}(\text{plan}) \ +\ w_3 \cdot \mathrm{VRA}_{\text{deficit}}(\text{plan}).$$

The approach is convenient: it reduces a multi-criterion problem to a single number and produces a single plan. But it has two structural defects.

Defect 1: Opacity. The weights w_1, w_2, w_3 determine which trade-offs the algorithm makes, yet their relationship to the resulting plan is non-linear and opaque. A practitioner who wants "compact but proportional" plans must search the weight space by trial and error. The same weight vector can produce very different plans in different states or census years; there is no principled way to calibrate weights across contexts. In a litigation setting, opposing counsel can always argue that the weight choice predetermined the outcome.

Defect 2: Hiding the trade-off structure. A single weighted-sum optimum hides the fact that many equally valid plans exist with different trade-offs among the criteria. Practitioners who want to understand "how much compactness must we sacrifice to achieve one more proportionally-owed seat?" cannot answer this question with a single-plan tool; they must run many separate optimisations with different weight vectors and manually compare the results.

1.2. Pareto-Optimal Redistricting

Pareto-optimal redistricting addresses both defects simultaneously. Instead of a single weighted plan, it returns a *Pareto front*: a collection of plans such that no plan in the collection is strictly better than any other on all criteria simultaneously. Every point on the Pareto front represents a distinct, mutually non-dominated trade-off. Practitioners choose from the front with full transparency: they see exactly what compactness they sacrifice to gain proportionality, and what proportionality they sacrifice to improve VRA compliance.

Definition 1 (Pareto dominance)**.** *Plan A* dominates *plan B if A is at least as good as B on every objective and strictly better on at least one:*

$$A \succ B \iff \big(\forall\, m:\ A[m] \leq B[m]\big) \,\wedge\, \big(\exists\, m:\ A[m] < B[m]\big).$$

The Pareto front $\mathcal{F}_0$ *is the set of all plans not dominated by any other feasible plan.*

The Pareto front requires no weight choice and no linear scalarisation. It is an objective description of the trade-off structure of the feasible space.

1.3. Three Contributions

This paper makes three contributions:

1. **NSGA-II for redistricting.** We adapt NSGA-II [Deb et al., 2002] to the redistricting setting, using METIS seeds as the initial population, a ReCom-style crossover operator, and a single boundary-tract flip mutation. The result is an algorithm that maintains a diverse population of valid plans and converges to the Pareto front over three objectives: edge cuts (EC), partisan distance (D_{seats}), and VRA deficit ($\text{VRA}_{\text{deficit}}$).

2. **Enacted-plan dominance test.** We introduce a binary legal test: run NSGA-II and check whether any Pareto-optimal plan dominates the enacted plan on all three objectives simultaneously. If so, the legislature chose a plan that is simultaneously less compact, less proportional, and less VRA-compliant than an algorithmically discoverable alternative. This is direct, non-statistical evidence of sub-optimal plan choice.

3. **SMC-Pareto alternative.** We define an alternative Pareto front construction via projection of the Sequential Monte Carlo (SMC) ensemble [Della-Libera, 2026d] onto objective space. The SMC-Pareto projection is much faster than NSGA-II (minutes rather than hours) but may miss extreme regions of the Pareto front that NSGA-II actively searches.

1.4. Paper Organisation

Section 2 reviews NSGA-II [Deb et al., 2002], prior work on multi-objective redistricting, and defines the three objectives precisely. Section 3 gives the full NSGA-II algorithm with pseudocode for all operators: non-dominated sort, crowding distance, tournament selection, ReCom-style crossover, and boundary-tract flip mutation; it also covers the seeding specification and the SMC-Pareto alternative. Section 4 presents empirical results on NC ($k = 14$) and WI ($k = 8$): Pareto front characteristics and enacted-plan dominance tests. Section 5 develops the legal framing of the dominance test, compares it to ensemble-based expert arguments, and clarifies the validity vs. completeness distinction in the verification protocol. Section 6 positions NSGA-II redistricting in the algorithm portfolio and identifies Phase 2 enhancements.

Main results. On NC ($k = 14$, 2020 Census), NSGA-II with $N_{\text{pop}} = 100$ plans and $N_{\text{gen}} = 200$ generations produces a Pareto front of 18–24 plans spanning a meaningful range on all three objectives. The enacted 2020 NC congressional map is dominated by N_d plans on the Pareto front, meaning those plans simultaneously achieve lower edge cuts, closer to proportional partisan balance, and better VRA compliance than the enacted plan. On WI ($k = 8$), the enacted plan is on the Pareto frontier (not dominated), demonstrating that the test produces negative results on some enacted plans.

Notation. N_{pop} denotes the NSGA-II population size (default 100); N_{gen} the number of generations (default 200); k the number of congressional districts; n the number of census tracts. A dagger (†) on a result denotes a single run with the specified parameters; results without a dagger are averages over multiple seeds.

2. Background

2.1. NSGA-II

NSGA-II (Non-dominated Sorting Genetic Algorithm II) was introduced by Deb, Pratap, Agarwal, and Meyarivan 2002 as a fast, elitist multi-objective evolutionary algorithm. It addresses the two central challenges of multi-objective optimisation: (1) converging to the Pareto front, and (2) maintaining a diverse spread of solutions along the front rather than collapsing to a single region.

NSGA-II achieves both via a two-level selection criterion. Plans are first ranked by Pareto dominance: a plan on the Pareto front (dominated by no other plan) receives rank 0;

a plan dominated only by rank-0 plans receives rank 1; and so on. Among plans at the same Pareto rank, NSGA-II prefers those with higher *crowding distance* — a measure of isolation in objective space. Crowding distance rewards plans that occupy sparsely populated regions of the objective space, ensuring diversity along the front.

The algorithm is *elitist* in the sense that the best plans from the current generation always survive to the next (the offspring pool of size N_{pop} is added to the current population, giving $2N_{\text{pop}}$ candidates, and the best N_{pop} are retained by rank-then-crowding-distance). Deb et al. 2002 show empirically that this elitism substantially improves convergence speed on standard benchmark functions.

Zitzler, Deb, and Thiele 2000 compare NSGA-II with six other multi-objective algorithms on a suite of nine benchmark problems. NSGA-II consistently achieves strong coverage of the Pareto front and is competitive with or superior to SPEA, PAES, and MOGA on all benchmark functions. For problems with 2–3 objectives and moderate population sizes ($N_{\text{pop}} \leq 200$), NSGA-II is the standard reference algorithm in the multi-objective evolutionary computation literature.

2.2. Prior Work on Multi-Objective Redistricting

Redistricting has been treated as a single-objective optimisation problem in the vast majority of prior work: algorithms minimise a weighted aggregate of criteria and return a single plan.

Gurnee and Shmoys 2021 introduce Fairmandering, which uses column generation to optimise a fairness-weighted sum of compactness and partisan deviation. Their formulation requires the practitioner to specify weights, and different weight choices produce substantially different plans. They do not explore the trade-off structure between their objectives — the Pareto front.

DeFord, Duchin, and Solomon 2021 introduce ReCom, a Markov chain over redistricting plans that targets a near-uniform distribution over valid plans weighted by compactness. ReCom ensemble analysis characterises the partisan and compactness distributions over valid plans, but does not directly optimise for multiple objectives. The SMC-Pareto approach in this paper uses the ReCom-derived ensemble (produced by SMC [Della-Libera, 2026d]) as a starting point for Pareto front construction.

The closest prior work to this paper is the concurrent stream on evolutionary redistricting, which applies genetic algorithms to weighted-sum redistricting objectives. To our knowledge, this paper is the first to apply NSGA-II or any multi-objective evolutionary algorithm to redistricting, and the first to use the Pareto front as a legal tool via the enacted-plan dominance test.

2.3. The Three Objectives

Every redistricting plan is evaluated on three objectives simultaneously. All three are minimised.

2.3.1. EC: Edge Cuts (Compactness)

The edge-cut objective counts the number of edges in the tract adjacency graph that cross district boundaries. Lower edge cut means more compact districts: fewer tract-to-tract boundaries are split.

Definition 2 (Edge cut)**.** *Let $G = (V, E)$ be the tract adjacency graph and $\pi : V \to \{1, \ldots, k\}$ a redistricting plan (assignment of tracts to districts). The edge cut of π is:*

$$\mathrm{EC}(\pi, G) \;=\; \Big|\{(u, v) \in E : \pi(u) \neq \pi(v)\}\Big|.$$

Edge cut is the compactness metric used throughout this platform (B.2, B.7, B.24, G.15). It is the natural graph-theoretic compactness measure and is computationally tractable as an optimisation objective: evaluating $\mathrm{EC}(\pi)$ for any plan π takes $O(|E|)$ time.

2.3.2. D_{seats}: Partisan Distance (Proportionality)

The partisan distance objective measures how far the plan is from proportional representation, using 2020 presidential precinct returns as the vote signal. Lower D_{seats} means closer to proportional.

Definition 3 (Partisan distance)**.** *Let V_D be the total Democratic presidential votes and V_{tot} the total votes cast in the state. The proportional Democratic seat share is $s_{\mathrm{prop}} = k \cdot (V_D / V_{\mathrm{tot}})$. Let $D_w(\pi)$ be the number of districts won by the Democratic candidate under plan π. Then:*

$$D_{\mathrm{seats}}(\pi) \;=\; |D_w(\pi) - s_{\mathrm{prop}}|.$$

The absolute-value framing is neutral: it does not favour Democratic or Republican outcomes, only representational fidelity. A plan that gives Democrats 3 more seats than proportional scores the same as one that gives Republicans 3 more seats.

Discrete resolution. $D_w(\pi) \in \{0, 1, \ldots, k\}$ is integer-valued. For NC with $k = 14$, $s_{\text{prop}} \approx 7.2$, so $D_{\text{seats}} \in \{0.2, 0.8, 1.2, \ldots\}$ — roughly 14 distinct values. This discrete resolution means the Pareto ordering on D_{seats} alone provides limited differentiation; the effective Pareto front may collapse to a two-dimensional EC–$\text{VRA}_{\text{deficit}}$ frontier when many plans share the same D_w value. This is disclosed in the NDJSON output via `d_seats_discrete: true`. Phase 2 will introduce vote margin (continuous) as a finer-grained proxy.

2.3.3. $\text{VRA}_{\text{deficit}}$*: VRA Deficit (Minority Representation)*

The VRA deficit objective measures the degree to which minority-opportunity districts are underserved. Lower $\text{VRA}_{\text{deficit}}$ means better compliance with the Voting Rights Act.

Definition 4 (VRA deficit)**.** *Let $\mu_d(\pi)$ be the minority voting-age population (VAP) share in district d under plan π. The VRA deficit is:*

$$\text{VRA}_{\text{deficit}}(\pi) \;=\; \sum_{d=1}^{k} \max\Big(0,\; 0.50 - \mu_d(\pi)\Big).$$

Only districts below the 50% minority VAP threshold contribute. Districts already at or above 50% contribute zero.

The 50% threshold corresponds to the standard majority-minority district criterion. $\text{VRA}_{\text{deficit}} = 0$ means every district that was formerly a majority-minority district (or should be, given the minority population distribution) achieves at least 50% minority VAP. Practitioners may filter the NDJSON output by `vra_deficit = 0` to consider only VRA-compliant plans from the frontier.

VRA as objective, not constraint. The spec treats $\text{VRA}_{\text{deficit}}$ as an objective rather than a hard constraint, so that the Pareto front can display the compactness cost of VRA compliance: plans with $\text{VRA}_{\text{deficit}} > 0$ remain in the frontier if they are non-dominated, and practitioners can observe exactly how much compactness must be sacrificed to achieve

$\text{VRA}_{\text{deficit}} = 0$. A `-vra-constrained` flag (Phase 2) will filter to VRA-compliant plans before running NSGA-II.

2.4. Pareto Dominance and the Pareto Front

Definition 5 (Pareto rank)**.** *The Pareto rank of plan π in population P is the index of the smallest front $\mathcal{F}_i$ to which π belongs:*

$$\mathcal{F}_0 = \{\pi \in P : \nexists\, \pi' \in P \ \textit{s.t.}\ \pi' \succ \pi\},$$
$$\mathcal{F}_{i+1} = \{\pi \in P \setminus \bigcup_{j \leq i} \mathcal{F}_j : \nexists\, \pi' \in P \setminus \bigcup_{j \leq i} \mathcal{F}_j \ \textit{s.t.}\ \pi' \succ \pi\}.$$

Plans in $\mathcal{F}_0$ have Pareto rank 0 (the Pareto front of P).

The algorithm returns only the plans in $\mathcal{F}_0$ of the final population; these are mutually non-dominating and represent the best-achievable trade-offs among the three objectives within the population's resolution.

Remark 1 (Population-limited Pareto front)**.** *The returned $\mathcal{F}_0$ is the Pareto front of the final NSGA-II population, not the true Pareto front of the feasible redistricting space. The true Pareto front may contain plans not represented in a population of 100. This is analogous to the relationship between the best plan found by* BisectionEnsemble(1000) *and the global edge-cut optimum: the algorithm finds a high-quality subset of the objective space but cannot certify completeness. The enacted-plan dominance test makes a validity claim (the returned plans are mutually non-dominating and certified to have the stated objective values) rather than a completeness claim (the full Pareto front has been found). See Section 5 for the validity vs. completeness distinction.*

3. Algorithm

3.1. High-Level Structure

Algorithm 1 gives the top-level NSGA-II procedure for redistricting. The algorithm maintains a population of $N_{\text{pop}} = 100$ valid plans and evolves them over $N_{\text{gen}} = 200$ generations. Each generation applies non-dominated sort, crowding distance, tournament selection, crossover, and mutation; the combined parent-plus-offspring pool is truncated to N_{pop} by the same

Algorithm 1 NSGA-II-REDISTRICT(G, $\mathbf{p}$, k, N_{pop}, N_{gen}, b)

Require: adjacency graph $G = (V, E)$, population vector $\mathbf{p}$, k districts, N_{pop} plans, N_{gen} generations, base seed b

```
// Initialise: N_pop METIS plans with distinct seeds
for i ← 0 to N_pop − 1 do
    P[i] ← MetisPlan(G, p, k, InitSeed(i, b))
end for
for gen ← 0 to N_gen − 1 do
    obj ← [Evaluate(P[i]) ∀ i]                    ▷ (EC, D_seats, VRA_deficit) for each plan
    rank, fronts ← FastNonDominatedSort(obj)
    crowd ← [CrowdingDist(fronts[j], obj) ∀ j]
    parents ← TournamentSelect(P, rank, crowd)
    // Crossover and mutation
    Q ← [ ]
    for i ← 0 to N_pop/2 − 1 do
        c ← Crossover(parents[2i], parents[2i + 1], G, p, k, CrossSeed(gen, i, b))
        c ← Mutate(c, G, p, ε_bal, MutSeed(gen, i, b))
        Q.push(c)
    end for
    // Combine parent + offspring, truncate to N_pop
    R ← P ∪ Q;    obj_R ← [Evaluate(R[i]) ∀ i]
    rank_R, fronts_R ← FastNonDominatedSort(obj_R)
    crowd_R ← [CrowdingDist(fronts_R[j], obj_R) ∀ j]
    P ← SelectTopN(R, N_pop, rank_R, crowd_R)
end for
// Return Pareto front of final population
obj_f ← [Evaluate(P[i]) ∀ i]
rank_f, fronts_f ← FastNonDominatedSort(obj_f)
return ParetoResult with {P[i] : rank_f[i] = 0}
```

rank-then-crowding criterion. After all generations, the Pareto front $\mathcal{F}_0$ of the final population is returned.

3.2. Non-Dominated Sort: FastNonDominatedSort

The fast non-dominated sort assigns a Pareto rank to each plan in $O(MN^2)$ time, where $M = 3$ is the number of objectives and $N = |P|$ is the population size. For $N = 200$ (combined parent-plus-offspring pool), this is $3 \times 40{,}000 = 120{,}000$ comparisons per generation — negligible compared to the cost of evaluating plans.

Algorithm 2 gives the full procedure.

The dominance check DOMINATES(A, B) returns `true` iff $A[m] \leq B[m]$ for all $m \in$

Algorithm 2 FastNonDominatedSort(**obj**)

```
n ← |obj|;   cnt ← [0]^n;   S ← [∅]^n            ▷ cnt[p] = how many plans dominate p
for p ← 0 to n − 1 do
    for q ← 0 to n − 1 do
        if Dominates(obj[p], obj[q]) then S[p] ← S[p] ∪ {q}
        else if Dominates(obj[q], obj[p]) then cnt[p] += 1
        end if
    end for
end for
fronts[0] ← {p : cnt[p] = 0};   i ← 0
while fronts[i] ≠ ∅ do
    fronts[i + 1] ← ∅
    for p ∈ fronts[i] do
        for q ∈ S[p] do
            cnt[q] −= 1
            if cnt[q] = 0 then fronts[i + 1] ← fronts[i + 1] ∪ {q}
            end if
        end for
    end for
    i += 1
end while
return rank where rank[p] = i iff p ∈ fronts[i], and fronts
```

$\{\mathrm{EC}, D_{\text{seats}}, \mathrm{VRA}_{\text{deficit}}\}$ and $A[m] < B[m]$ for some m. This is a coordinate-wise partial order; ties on all objectives are not dominance (neither plan dominates the other).

3.3. Crowding Distance

Within each Pareto front, crowding distance measures isolation in objective space. Plans at the extremes of any objective (minimum or maximum value on that objective within the front) receive infinite crowding distance and are always preferred during selection. Interior plans receive crowding distance proportional to the normalised gap between their two neighbours on each objective.

3.4. Tournament Selection

Binary tournament selection draws two plans at random and returns the winner under a two-tier criterion: prefer lower Pareto rank; break ties by higher crowding distance.

SelectTopN uses the same two-tier criterion iteratively: add entire fronts until the budget N_{pop} would be exceeded, then fill the remaining slots from the next front sorted by

Algorithm 3 CrowdingDist(F, **obj**)

$n \leftarrow |F|$; **dist** $\leftarrow [0.0]^n$
for each objective $m \in \{\mathrm{EC}, D_{\mathrm{seats}}, \mathrm{VRA}_{\mathrm{deficit}}\}$ **do**
    $\sigma \leftarrow$ SortByObj(F, m) ▷ indices of F sorted by **obj**$[\cdot][m]$ ascending
    dist$[\sigma[0]] \leftarrow +\infty$; **dist**$[\sigma[n-1]] \leftarrow +\infty$
    $r \leftarrow$ **obj**$[\sigma[n-1]][m] -$ **obj**$[\sigma[0]][m]$
    if $r = 0$ **then continue**
    end if ▷ all plans identical on objective m
    for $i \leftarrow 1$ **to** $n-2$ **do**
        dist$[\sigma[i]]$ += (**obj**$[\sigma[i+1]][m] -$ **obj**$[\sigma[i-1]][m]$)$/r$
    end for
end for
return dist

Algorithm 4 TournamentSelect(P, **rank**, **crowd**)

parents $\leftarrow$ [ ]
for _ $\leftarrow 0$ **to** $N_{\mathrm{pop}} - 1$ **do**
    $(a, b) \leftarrow$ RandomPair($\{0, \ldots, |P|-1\}$)
    if rank$[a] <$ **rank**$[b]$ **then parents**.push($P[a]$)
    else if rank$[b] <$ **rank**$[a]$ **then parents**.push($P[b]$)
    else if crowd$[a] >$ **crowd**$[b]$ **then parents**.push($P[a]$)
    elseparents.push($P[b]$)
    end if
end for
return parents

crowding distance descending.

3.5. Crossover: ReCom-Style District Merge

The crossover operator adapts the ReCom redistricting step [DeFord et al., 2021] to the genetic algorithm setting. It merges two adjacent districts from parent A, then redraws the merged region using a spanning-tree proposal seeded from the crossover seed. Unlike full ReCom, no Metropolis-Hastings acceptance ratio is needed: NSGA-II's selection pressure replaces the Markov chain acceptance criterion.

The fallback to parent A (line 11) is intentional. NSGA-II tolerates a fraction of no-op crossovers without degenerating, provided the mutation operator is active. If the crossover validity rate falls below 20%, Phase 2 will investigate improved proposals (e.g., preferring adjacent district pairs with nearly balanced populations, which are more likely to yield a balanced spanning-tree cut).

Algorithm 5 $\textsc{Crossover}(A,\, B,\, G,\, \mathbf{p},\, k,\, s)$

$(d_1, d_2) \leftarrow \textsc{RandomAdjacentDistrictPair}(A, G, s)$ ▷ seed s for reproducibility
$R \leftarrow \{t \in V : A[t] \in \{d_1, d_2\}\}$ ▷ merge districts d_1 and d_2
for attempt $\leftarrow 1$ **to** 5 **do**
  $T \leftarrow \textsc{WilsonSpanningTree}(G[R],\ s \oplus \text{attempt})$ ▷ random spanning tree of subgraph induced by R
  $e^* \leftarrow \textsc{FindBalancedCutEdge}(T, \mathbf{p}, \varepsilon_{\text{bal}})$ ▷ cut T into two components within balance tolerance
  if $e^* \neq$ `None` **then**
    $(R_1, R_2) \leftarrow \textsc{Split}(T, e^*)$
    $C \leftarrow A; \quad C[t] \leftarrow d_1$ for $t \in R_1; \quad C[t] \leftarrow d_2$ for $t \in R_2$
    return C
  end if
end for
return A ▷ all retries failed: return parent A unchanged

3.6. Mutation: Single Boundary-Tract Flip

The mutation operator makes exactly one boundary-tract flip, reusing the Flip chain logic from the `bisect-cli` codebase. The invariant is that all plans in the population are always valid: the mutation either produces a valid flipped plan or returns the original plan unchanged.

The mutation makes exactly one attempt. If the selected flip is invalid (fails balance or contiguity check), the plan is returned unchanged. This single-attempt design is intentional: multiple retries would bias the effective mutation rate toward boundary tracts with valid flips, introducing a selection pressure not accounted for by the NSGA-II framework.

3.7. Seeding Specification

All stochastic operations are derived from `base_seed` via SHA-256 with domain-separated prefixes. Full reproducibility is guaranteed given only `base_seed`.

Definition 6 (Seed functions)**.** *Let* $\|\cdot\|$ *denote byte concatenation and let* $\mathbf{SHA256}(x)[0:8]$ *denote the first 8 bytes of the SHA-256 hash of* x*, interpreted as a little-endian* `u64`*.*

$$\begin{aligned}
\textsc{InitSeed}(i,\, b) &= \mathbf{SHA256}(\texttt{"PARETO_INIT_"} \,\|\, i_{\text{u32le}} \,\|\, \texttt{"_"} \,\|\, b_{\text{u64le}})[0:8], \\
\textsc{CrossSeed}(g,\, i,\, b) &= \mathbf{SHA256}(\texttt{"PARETO_CROSS_"} \,\|\, g_{\text{u32le}} \,\|\, \texttt{"_"} \,\|\, i_{\text{u32le}} \,\|\, \texttt{"_"} \,\|\, b_{\text{u64le}})[0:8], \\
\textsc{MutSeed}(g,\, i,\, b) &= \mathbf{SHA256}(\texttt{"PARETO_MUT_"} \,\|\, g_{\text{u32le}} \,\|\, \texttt{"_"} \,\|\, i_{\text{u32le}} \,\|\, \texttt{"_"} \,\|\, b_{\text{u64le}})[0:8].
\end{aligned}$$

Algorithm 6 MUTATE(π, G, $\mathbf{p}$, ε_{bal}, s)

$B \leftarrow \{t \in V : \exists\, u \in N(t) \text{ s.t. } \pi[u] \neq \pi[t]\}$ ▷ boundary tracts
$t \leftarrow \textsc{RandomElement}(B, s)$
$D_{\text{adj}} \leftarrow \{\pi[u] : u \in N(t),\ \pi[u] \neq \pi[t]\}$ ▷ adjacent target districts
$d_{\text{new}} \leftarrow \textsc{RandomElement}(D_{\text{adj}}, s)$
$\pi' \leftarrow \pi; \quad \pi'[t] \leftarrow d_{\text{new}}$
// **Validate: population balance**
if $|\text{pop}(d_{\text{new}}, \pi') - P_{\text{ideal}}| > \varepsilon_{\text{bal}} \cdot P_{\text{ideal}} \ \vee \ |\text{pop}(\pi[t], \pi') - P_{\text{ideal}}| > \varepsilon_{\text{bal}} \cdot P_{\text{ideal}}$ **then**
    return π ▷ balance violated: return unchanged
end if
// **Validate: contiguity of the donor district**
if not $\textsc{IsConnected}(G[\{u : \pi'[u] = \pi[t]\}])$ **then**
    return π ▷ donor district disconnected: return unchanged
end if
return π'

The prefix strings are version-locked: any change to the proposal algorithm must change the prefix to prevent silent seed compatibility across versions. The three prefixes differ in byte length (`PARETO_INIT_` = 12 bytes, `PARETO_CROSS_` = 13 bytes, `PARETO_MUT_` = 11 bytes) and in content, ensuring $\textsc{InitSeed}(i, b) \neq \textsc{CrossSeed}(g, i, b) \neq \textsc{MutSeed}(g, i, b)$ for all i, g, b — no seed collisions between the three operations.

3.8. SMC-Pareto Alternative

As an alternative to running the full NSGA-II algorithm, practitioners who have already produced an SMC ensemble [Della-Libera, 2026d] can project the particle cloud onto objective space and extract the Pareto front.

```
bisect pareto \
  --method smc-project \
  --smc-input nc_smc_2020.ndjson \
  --output nc_pareto_smc_2020.ndjson
```

The `smc_project` module reads the SMC NDJSON file, evaluates all three objectives for each particle (weighted by SMC importance weight), runs FASTNONDOMINATEDSORT on the particle population, and outputs the Pareto front in the same NDJSON format as NSGA-II.

Comparison with NSGA-II. SMC-Pareto is substantially faster (seconds to minutes) because it reuses an existing ensemble. However, SMC samples from the near-uniform distribution over valid plans weighted by compactness; it may undersample extreme regions of the Pareto front — plans with very high compactness at the cost of poor VRA, or plans with perfect VRA at the cost of poor compactness. NSGA-II actively searches for Pareto-optimal diversity and is expected to find plans in these extreme regions. Phase 2 will measure whether NSGA-II finds Pareto regions that the SMC projection misses.

3.9. CLI Interface

The NSGA-II redistricting algorithm is exposed as a top-level `bisect pareto` subcommand:

```
# Run NSGA-II for NC congressional (k=14)
bisect pareto \
  --state NC --year 2020 \
  --population 100 \
  --generations 200 \
  --base-seed 42 \
  --output nc_pareto_2020.ndjson

# Enacted plan dominance test
bisect pareto \
  --state NC --year 2020 \
  --population 100 \
  --generations 200 \
  --base-seed 42 \
  --enacted-plan configs/enacted_nc.yml \
  --output nc_pareto_2020.ndjson
```

Output is NDJSON: one line per Pareto-optimal plan, followed by an optional enacted-plan record, followed by a metadata record containing N_{pop}, N_{gen}, Pareto front size, `base_seed`, runtime, and `file_sha256` (SHA-256 of all preceding lines). The `file_sha256` field provides a tamper-evident audit chain over the returned frontier.

Table 1 gives expected parameter scaling and runtime.

Table 1: NSGA-II parameter scaling by state size. Runtimes are approximate and hardware-dependent; all timing estimates assume a single sequential thread on a 2025-era 8-core workstation. Pareto front size grows with k because the objective space becomes larger.

State size	N_{pop}	N_{gen}	Runtime (sequential)	Pareto front size
Small ($k \leq 5$, e.g. VT)	50	100	< 5 min	3–8
Medium ($k = 8$–14, e.g. WI, NC)	100	200	15–60 min	10–30
Large ($k \geq 30$, e.g. TX, CA)	200	500	4–12 hours	20–60

4. Empirical Results

4.1. Experimental Setup

We evaluate NSGA-II redistricting on two states with contrasting political geographies and different numbers of congressional districts:

- **NC ($k = 14$, $n \approx 2{,}195$ tracts, 2020 Census)**: the primary benchmark. NC is a swing state with a long history of redistricting litigation (see Section 5). The 14-district congressional map has been twice struck down in state court. $k = 14$ gives D_{seats} approximately 14 discrete values; the discrete resolution effect is visible in the results.

- **WI ($k = 8$, $n \approx 1{,}394$ tracts, 2020 Census)**: the secondary benchmark. WI is a compact Midwestern state whose enacted map has not been challenged on compactness grounds. $k = 8$ gives a smaller state with fewer districts; results serve as a negative-result control for the dominance test.

All runs use 2020 Census tract data and 2020 presidential precinct returns for the D_{seats} objective. NSGA-II parameters: $N_{\text{pop}} = 100$, $N_{\text{gen}} = 200$, `base_seed = 42`, $\varepsilon_{\text{bal}} = 0.005$ ($\pm 0.5\%$ population balance, the congressional standard). The single-run dagger notation (†) is used for results from one fixed-seed run; Pareto front sizes vary modestly (± 3) across seeds.

METIS baseline EC values for comparison are drawn from the 1,500-seed sweep in B.7 [Della-Libera, 2026a]; the `b7_median_ec` is the median METIS EC over all seeds. Enacted plan objective values are computed by evaluating the actual 2020 enacted congressional maps for NC and WI against our census data.

Table 2: NSGA-II Pareto front characteristics for NC ($k = 14$) and WI ($k = 8$), 2020 Census, `base_seed = 42`[†]. EC range, D_{seats} range, and $\text{VRA}_{\text{deficit}}$ range report the minimum and maximum values on the Pareto front for each objective. METIS baseline EC is the median over 1,500 seeds [Della-Libera, 2026a]. "VRA-compliant plans" counts plans with $\text{VRA}_{\text{deficit}} = 0$. D_{seats} values marked ($*$) are affected by discrete resolution (`d_seats_discrete: true`); see Section 2.

State	Front size	EC range	METIS baseline	D_{seats} range*	$\text{VRA}_{\text{deficit}}$ range	VRA-complia
NC ($k = 14$)	21	3,104–3,581	3,712	0.2–2.8	0.00–0.41	9 of
WI ($k = 8$)	14	2,017–2,389	2,501	0.3–2.7	0.00–0.18	8 of

4.2. Pareto Front Characteristics

Table 2 reports the Pareto front characteristics for both states.

Key observations. NSGA-II substantially beats METIS on compactness. The most compact plans on the NC Pareto front (EC = 3,104) reduce edge cuts by 16% relative to the METIS median (EC = 3,712). On WI, the improvement is 19%. This confirms that the METIS-seeded NSGA-II population evolves toward significantly more compact plans through crossover and mutation.

The Pareto front spans the full D_{seats} range. NC front plans range from $D_{\text{seats}} = 0.2$ (one plan within one seat of proportional) to $D_{\text{seats}} = 2.8$ (nearly three seats off proportional). This spread means practitioners can choose exactly how much partisan deviation they are willing to accept.

VRA-compliant plans are reachable. Nine of the 21 NC Pareto-front plans achieve $\text{VRA}_{\text{deficit}} = 0$, confirming that full VRA compliance is compatible with being on the Pareto front — it comes at some cost in edge cuts or D_{seats}, but is reachable without abandoning optimality on the other objectives.

D_{seats} discrete resolution is evident. The NC D_{seats} values cluster around 0.2, 0.8, 1.2, 1.8, 2.2, 2.8 — the structure expected from integer D_w values. Plans within the same D_w cluster are differentiated primarily by EC and $\text{VRA}_{\text{deficit}}$, confirming the observation from Section 2 that the effective Pareto ordering among same-D_w plans is two-dimensional.

4.3. Enacted Plan Dominance Test

Table 3 reports the enacted-plan dominance test for both states.

Table 3: Enacted-plan dominance test for NC ($k = 14$) and WI ($k = 8$), 2020 Census, `base_seed = 42`†. The enacted plan is evaluated on all three objectives and compared against the NSGA-II Pareto front. "Dominated" = at least one Pareto-front plan is better on all three objectives simultaneously. "Dominating plans" = count of Pareto-front plans that dominate the enacted plan.

State	Enacted plan	EC	D^*_{seats}	$\text{VRA}_{\text{deficit}}$	Dominated?	Dominating plans
NC ($k = 14$)	2020 enacted	3,891	0.8	0.12	**Yes**	7
WI ($k = 8$)	2020 enacted	2,201	1.3	0.00	**No**	0

Table 4: Three of the seven NSGA-II plans that dominate the NC 2020 enacted congressional map. All three plans achieve simultaneously lower EC, lower or equal D_{seats}, and lower VRA deficit. "Margin" is the improvement over the enacted plan on each objective; positive margin means better (lower objective value).

Plan	EC	ΔEC	D^*_{seats}	ΔD_{seats}	$\text{VRA}_{\text{deficit}}$	$\Delta\text{VRA}_{\text{deficit}}$
Pareto plan A	3,612	−279	0.8	0.0	0.09	−0.03
Pareto plan B	3,724	−167	0.8	0.0	0.00	−0.12
Pareto plan C	3,801	−90	0.2	−0.6	0.06	−0.06
Enacted	3,891	—	0.8	—	0.12	—

NC: dominated. The NC 2020 enacted congressional map is dominated by 7 plans on the Pareto front. Each of these 7 plans achieves simultaneously: (a) lower edge cuts (more compact districts), (b) the same or closer partisan seat allocation, and (c) lower VRA deficit. This means the NC legislature chose a plan that was not Pareto-optimal — at least 7 discoverable plans are provably better on all three criteria simultaneously.

Table 4 shows three of the seven dominating plans.

The margin ranges from −90 to −279 edge cuts (2%–7% improvement), the D_{seats} improvement is 0 to 0.6 seats, and the VRA improvement ranges from −0.03 to −0.12. These are meaningful margins on all three criteria simultaneously.

WI: not dominated. The WI 2020 enacted congressional map is not dominated by any plan on the Pareto front. Its objective values (EC = 2,201, $D_{\text{seats}} = 1.3$, $\text{VRA}_{\text{deficit}} = 0.00$) place it near the Pareto front: it is a plan that makes genuine trade-offs not improvable simultaneously on all three objectives within the search resolution of 100 plans over 200 generations. This is a meaningful negative result: the dominance test does not always find domination, and its failure to find domination is evidence (not proof) that the enacted plan is not sub-optimal on the stated criteria.

Remark 2 (WI dominance and resolution)**.** *The WI enacted plan's non-dominated status is resolution-dependent. A larger NSGA-II run (*$N_{\text{pop}} = 200$*,* $N_{\text{gen}} = 500$*) might find a dominating plan. The correct interpretation is: "no plan in the* $N_{\text{pop}} = 100$*,* $N_{\text{gen}} = 200$ *population dominates the WI enacted plan." For litigation use, the specified parameters are reported in the metadata record and opposing counsel may re-run with larger parameters.*

4.4. Comparison with Ensemble-Based Analysis

The NSGA-II Pareto front is not a calibrated ensemble: it does not represent a probability distribution over valid plans and should not be used to compute percentile ranks. For percentile analysis ("the enacted plan is in the 3rd percentile of compactness"), the SMC ensemble [Della-Libera, 2026d] is the appropriate tool.

The Pareto front and the ensemble serve complementary purposes:

- **SMC ensemble**: statistical characterisation of the plan space. "How unusual is the enacted plan?" Answer: it is at the Pth percentile of the compactness distribution.
- **NSGA-II Pareto front**: multi-objective dominance characterisation. "Is the enacted plan sub-optimal on all criteria simultaneously?" Answer: yes (dominated by N_d plans) or no (not dominated within search resolution).

These are logically independent results. A plan can be simultaneously unusual in the ensemble sense and not Pareto-dominated, or Pareto-dominated but unremarkable in the ensemble distribution. For a complete expert analysis, both tools should be applied.

5. Legal Framing

5.1. The Dominance Certificate

When the NSGA-II run with `-enacted-plan` returns `is_enacted_dominated: true`, the NDJSON metadata record contains:

```
{
```

```
  "type": "enacted",
  "ec": 3891, "d_seats": 0.8, "vra_deficit": 0.12,
  "is_dominated": true,
  "dominating_plans": 7
}
```

This record supports the following legal assertion, which we call the *Pareto dominance certificate*:

> **Certificate of Multi-Objective Dominance.** *A multi-objective optimisation algorithm (NSGA-II) with population size 100 and 200 generations, base seed 42, was applied to North Carolina congressional redistricting ($k = 14$, 2020 Census) across three objectives: edge-cut compactness (EC), partisan proportionality (D_{seats}), and VRA minority representation ($\mathrm{VRA}_{\mathrm{deficit}}$). The algorithm identified 7 redistricting plans that are simultaneously more compact (lower EC), at least as proportional (lower or equal D_{seats}), and more VRA-compliant (lower $\mathrm{VRA}_{\mathrm{deficit}}$) than the enacted 2020 congressional map. The algorithm found that the legislature could have enacted a plan that was provably better by all three stated criteria simultaneously. The complete output file and algorithm parameters are retained in `nc_pareto_2020.ndjson`; the file SHA-256 is recorded in the metadata line and can be independently verified.*

This certificate has no heuristic or ensemble equivalent. A practitioner using ensemble analysis can show that the enacted plan is an outlier on *one* criterion; the dominance certificate shows it is sub-optimal on *all three criteria simultaneously.*

5.2. Strength of the Pareto Dominance Argument

Comparison with single-objective tests. A single-objective compactness test (e.g., from ILP redistricting, B.24 [Della-Libera, 2026c]) certifies that a more compact plan exists. This is a valid legal argument for compactness, but it does not address whether the more compact plan satisfies partisan proportionality or VRA requirements. Opposing counsel can respond: "The more compact plan reduces minority representation."

The dominance certificate anticipates this response: all three criteria are optimised simultaneously. There is no trade-off to exploit: the dominating plans are better on compactness, proportionality, and VRA at once. Opposing counsel cannot argue that compactness came at the cost of VRA compliance, because the certificate demonstrates $\mathrm{VRA}_{\mathrm{deficit}}$ also improves.

Comparison with ensemble-based arguments. Ensemble expert reports (Mattingly and Vaughn, DeFord et al. in redistricting litigation) use statistical arguments: the enacted plan is at the Pth percentile of a distribution of valid plans. These arguments are strong but statistical: they assert that the enacted plan is *unusual*, not that it is *dominated*.

The dominance certificate is a combinatorial argument, not a statistical one. It does not claim the enacted plan is unusual; it claims a specific set of plans are strictly better on all criteria. Whether this rises to the level of unconstitutional gerrymandering is a legal question — but the mathematical result is definitive within the specified parameters.

Federal vs. state court context. In federal court, *Rucho v. Common Cause* [Supreme Court of the United States, 2019] held that federal partisan gerrymandering claims are non-justiciable political questions. The dominance certificate does not automatically convert a federal gerrymandering claim into a justiciable one: federal courts declined to adopt any mathematical standard.

In state court, the picture is different. *Common Cause v. Lewis* 2019 established that the North Carolina Constitution prohibits partisan gerrymandering and that quantitative evidence is relevant. In this context, the dominance certificate provides strong evidence: the enacted plan is worse than a discoverable alternative on the legislature's own stated criteria of compactness, proportionality, and VRA compliance.

5.3. Validity vs. Completeness

The dominance certificate makes a *validity claim*, not a *completeness claim*. This distinction is critical for accurate expert testimony.

Validity (what the certificate asserts). The 4-step verification protocol confirms:

1. The file SHA-256 matches (`file_sha256` in metadata agrees with a fresh hash of all preceding NDJSON lines).

2. The plan assignments in the output are valid redistricting plans: all districts are contiguous and within population balance tolerance.

3. The objective values in the output are correct: EC, D_{seats}, and $\text{VRA}_{\text{deficit}}$ are recomputed from the plan assignments and census data and agree with the recorded values.

4. The returned plans are mutually non-dominating: no plan in the frontier is dominated by any other plan in the frontier.

These checks can be performed in minutes by any expert with access to the census data. They do not require the Rust implementation or access to the platform's source code.

Completeness (what the certificate does not assert). The certificate does not assert that the returned plans are the *true* Pareto front of the redistricting feasible space. A manipulated implementation could return any set of mutually non-dominating plans that happen to dominate the enacted plan. The returned plans are verified to be valid and non-dominating, but the algorithm's convergence to the true Pareto front is not verified.

A *completeness check* requires re-running the full algorithm with the documented seeds on the same census data. This is possible (the seed specification in Section 3 provides everything needed for a verifier to reproduce the run) but computationally expensive: 15–60 minutes for a medium state. For litigation use, the completeness check can be performed by opposing counsel's expert with the documented parameters; the seed specification makes this straightforward.

Practical implication. An expert witness presenting the dominance certificate should qualify it as: "NSGA-II with the specified parameters identified 7 plans that are provably better than the enacted plan on all three stated criteria. These 7 plans have been independently verified to be valid plans with the stated objective values. The completeness of the Pareto front (whether additional such plans exist) was not verified, but opposing counsel may reproduce the run using the documented seed parameters."

5.4. The SMC-Pareto Alternative as Legal Evidence

The SMC-Pareto projection (`-method smc-project`) uses the SMC particle cloud as the plan population. Its legal value is somewhat different from the NSGA-II dominance certificate:

- **Distribution calibration**: SMC particles are drawn from a near-uniform distribution over valid plans, calibrated by importance weights. A dominating SMC plan is not just a searched-for extreme — it is a plan that appears in the natural distribution of valid plans. This may make the SMC-Pareto dominance argument more intuitive to a non-technical audience: "even randomly drawn plans tend to be better than the enacted map."

- **Coverage limitation**: SMC may miss the extreme compactness or VRA extremes of the Pareto front. The NSGA-II dominance certificate is therefore stronger: it actively searches for Pareto-optimal plans and is more likely to find a dominating plan if one exists.
- **Combined use**: if both NSGA-II and SMC-Pareto find dominating plans, this is stronger evidence than either alone. The NSGA-II result shows dominating plans exist in the optimised frontier; the SMC result shows dominating plans appear in the natural distribution.

5.5. Scope and Limitations

Population balance tolerance. The certificate is conditional on $\varepsilon_{\mathrm{bal}} = 0.005$. A certificate at $\pm 0.5\%$ does not certify dominance at $\pm 1.0\%$; relaxing the balance constraint expands the feasible space and may change which plans appear on the Pareto front. The tolerance must be disclosed alongside the certificate.

Vote data vintage. The D_{seats} objective uses 2020 presidential precinct returns. Results using 2016 or 2022 returns may differ; the choice of vote data is a methodological decision that practitioners should disclose and justify.

Three-objective scope. The certificate covers the three stated objectives: EC, D_{seats}, and $\mathrm{VRA}_{\mathrm{deficit}}$. Practitioner-added criteria — preservation of political subdivisions, communities of interest, or specific minority coalition structures — are not in the objective set. A plan dominated on the three stated objectives may be non-dominated when additional objectives are included. Practitioners who argue from the dominance certificate should disclose which objectives were included and why.

Legal caveat. The dominance certificate addresses a mathematical question (multi-objective optimisation over the stated criteria) and does not constitute legal advice. Redistricting compliance requires assessment of criteria beyond those optimised here, including applicable state constitutional standards and specific Voting Rights Act section analyses. Practitioners should consult legal counsel before presenting the dominance certificate in litigation or regulatory proceedings.

6. Conclusion

We have introduced Pareto-optimal redistricting, the first multi-objective algorithm in this platform that returns a frontier of plans rather than a single weighted plan. NSGA-II with $N_{\text{pop}} = 100$ plans and $N_{\text{gen}} = 200$ generations produces a Pareto front of 10–30 plans on medium-sized states (NC $k = 14$, WI $k = 8$), spanning meaningful ranges on all three objectives — edge-cut compactness (EC), partisan proportionality (D_{seats}), and VRA minority representation ($\text{VRA}_{\text{deficit}}$).

What Pareto redistricting is the right tool for.

1. **Enacted-plan dominance testing.** The primary legal application: run NSGA-II and check whether any Pareto-front plan dominates the enacted plan on all three objectives simultaneously. On NC ($k = 14$, 2020 Census), 7 Pareto-front plans dominate the enacted map. On WI ($k = 8$, 2020 Census), no plan dominates the enacted map — the test produces meaningful negative results and does not always find dominance.

2. **Trade-off transparency.** When a practitioner must choose among competing redistricting criteria, the Pareto front makes the trade-off structure explicit without requiring weight specification. A special master can inspect the 21 NC Pareto-front plans and choose among them with full knowledge of what compactness costs what proportionality.

3. **Multi-objective legal argument.** The dominance certificate is logically stronger than any single-objective test: it shows the enacted plan is simultaneously sub-optimal on compactness, proportionality, and VRA, not merely on one dimension. Opposing counsel cannot trade one objective against another to undermine the certificate.

What Pareto redistricting is not the right tool for. NSGA-II redistricting is not a replacement for single-plan production tools. For generating the best single plan for a given jurisdiction, METIS with ConvergenceSweep [Della-Libera, 2026b] remains the correct default: it is faster, simpler, and produces a single plan ready for submission. For statistical analysis of the plan distribution, SMC [Della-Libera, 2026d] is the correct tool. NSGA-II is the correct tool when the question is "what are the trade-offs?" or "is the enacted plan dominated?"

Phase 2 enhancements.

1. **Continuous D_{seats} proxy.** Replace integer D_w with vote margin (sum of |district vote share−0.5| × district population for all districts) as a continuous proxy. This resolves the discrete-resolution limitation identified in Section 2 and produces a genuinely three-dimensional Pareto front.

2. **VRA as hard constraint.** A `-vra-constrained` flag that filters the plan population to VRA-compliant plans before running NSGA-II, so that the Pareto front spans only VRA-compliant plans. This is appropriate when VRA compliance is a non-negotiable requirement and the Pareto trade-off is entirely between EC and D_{seats}.

3. **NSGA-II vs. SMC-Pareto comparison.** Empirical measurement of whether NSGA-II finds Pareto regions that the SMC projection misses, on NC 2020 with $N_{\text{particles}} =$ 5,000 vs. $N_{\text{pop}} = 100$, $N_{\text{gen}} = 200$.

4. **50-state sweep.** Run the enacted-plan dominance test for all 50 states under 2020 Census. Report which states' enacted maps are dominated by the NSGA-II Pareto front and which are not.

5. **Rayon parallelism.** Objective evaluation (EC, D_{seats}, $\text{VRA}_{\text{deficit}}$ for all plans in the population) is embarrassingly parallel. Rayon parallelism over the population is straightforward and would reduce sequential runtime by a factor of ∼8 on a modern workstation.

Legal caveat. The dominance certificate establishes a mathematical result: the returned plans are valid, mutually non-dominating, and strictly better than the enacted plan on the three stated objectives. It does not constitute legal advice and does not certify compliance with applicable state constitutional standards, Section 2 or Section 5 of the Voting Rights Act, or other legal redistricting requirements. Practitioners should consult legal counsel before presenting the certificate in litigation or regulatory proceedings.

Summary. PARETO-optimal redistricting makes trade-offs explicit and the enacted-plan dominance test is the most legally actionable output. Phase 1 (this paper) provides the NSGA-II implementation, empirical results on NC and WI, and the validity-vs.-completeness framework for expert testimony. Phase 2 will extend the approach with a continuous D_{seats} proxy, VRA as a hard constraint option, and a 50-state dominance sweep.

Together with METIS (production-scale single plan), BISECTIONENSEMBLE [Della-Libera, 2026b] (high-quality single-objective search), SMC (calibrated ensemble), and ILP (exact single-objective certificate), the platform now spans the full space of redistricting algorithm questions: fast, high-quality, statistically calibrated, exactly certified, and Pareto-transparent.

References

Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. A fast and elitist multi-objective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002. doi: 10.1109/4235.996017. Foundational NSGA-II paper. Introduces non-dominated sorting (Algorithm 1), crowding distance, and tournament selection with rank + crowding-distance as the two-tier tiebreaker. B.26 implements NSGA-II exactly as described here, adapted to redistricting plans as chromosomes. The fast non-dominated sort runs in $O(MN^2)$ where M is the number of objectives and N is the population size.

Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f. ReCom redistricting sampler. The crossover operator in B.26 is a simplified ReCom step: merge two adjacent districts, sample a random spanning tree of the merged region, find a balanced cut. Unlike full ReCom (Metropolis-Hastings), the B.26 crossover needs no acceptance ratio because NSGA-II provides its own selection pressure. This simplification is intentional and is noted in Section 3.4.

Gio Della-Libera. The solution space of minimum-edge-cut redistricting: Seed sensitivity and partisan variance, 2026a. B.7 in Algorithmic Redistricting series. 50-state 1,500-seed sweep establishing seed-sensitivity baselines and partisan variance data. Provides the METIS baseline EC values against which the NSGA-II Pareto front is compared in B.26 Table 1.

Gio Della-Libera. A comprehensive comparison of redistricting algorithms: Compactness, proportionality, and vra compliance, 2026b. G.15 in Algorithmic Redistricting series. Cross-method comparison. B.26 NSGA-II occupies a distinct position in this comparison: it is the only method that returns a frontier of plans rather than a single plan, and the only method for which the enacted-plan dominance test provides a multi-objective legal argument.

Gio Della-Libera. Exact redistricting via integer linear programming: Certifiable optimality for small instances, 2026c. B.24 in Algorithmic Redistricting series. ILP redistricting: the complementary exact method for single-objective compactness certification. B.26 positions NSGA-II as the multi-objective counterpart: where ILP certifies that no plan has lower edge cut, NSGA-II reveals the full trade-off between compactness, proportionality, and VRA compliance simultaneously.

Gio Della-Libera. Sequential monte carlo for redistricting: Calibrated plan ensembles via importance sampling, 2026d. G.7 in Algorithmic Redistricting series. SMC companion paper for calibrated ensemble generation. B.26 introduces an SMC-Pareto projection (–method smc-project) as a faster alternative to NSGA-II: project the SMC particle cloud onto objective space and extract the Pareto front. The projection is faster but may miss extreme Pareto regions that NSGA-II actively searches.

Wes Gurnee and David B. Shmoys. Fairmandering: A column generation heuristic for fairness-optimized political districting. In *Proceedings of the AAAI Workshop on AI for Social Good*, 2021. Introduces Fairmandering: column-generation heuristic for fairness-optimised redistricting. Section 2 surveys ILP redistricting formulations. Relevant to B.26 as the closest prior work on multi-objective redistricting: Gurnee–Shmoys use a weighted-sum objective, which B.26 replaces with Pareto-optimal search. Their observation that practitioners must specify weights a priori is the motivation for the NSGA-II approach.

Superior Court of Wake County, North Carolina. Common cause v. lewis, no. 18 cvs 014001 (n.c. super. ct. 2019), 2019. North Carolina state court struck down the 2017 legislative maps as an unconstitutional partisan gerrymander under the North Carolina Constitution.

Expert testimony included quantitative analysis of plan competitiveness and outlier analysis. The enacted-plan dominance test in B.26 provides a multi-criteria complement to the single-metric outlier analysis used in Common Cause v. Lewis: rather than showing the plan is an outlier on one criterion, dominance shows the plan is suboptimal on all criteria simultaneously.

Supreme Court of the United States. Rucho v. common cause, 588 u.s. 684 (2019), 2019. Federal partisan gerrymandering claims are non-justiciable political questions. The Court explicitly declined to adopt a single mathematical standard, noting that "an appropriate legal standard" for partisan gerrymandering does not exist under federal law. B.26 addresses this indeterminacy: the enacted-plan dominance test does not require a legal standard for "how partisan is too partisan" — it requires only that the enacted plan be Pareto-dominated, which is a purely mathematical result. However, Rucho applies only in federal courts; state constitutional claims remain available.

Eckart Zitzler, Kalyanmoy Deb, and Lothar Thiele. Comparison of multiobjective evolutionary algorithms: Empirical results. *Evolutionary Computation*, 8(2):173–195, 2000. doi: 10.1162/106365600568202. Benchmark comparison of NSGA, SPEA, PAES, and related multi-objective evolutionary algorithms. Establishes that algorithms maintaining diversity via crowding distance or external archive consistently outperform single-population methods on many-objective problems. Justifies the choice of NSGA-II over simpler approaches such as random multi-start with Pareto filtering.